

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

(a) 計算判定係數 R^2

1. 計算總平方和 (SST)

$$SST = \sum y_i^2 - N\bar{y}^2$$

$$SST = 7825.94 - 20 \times (19.21)^2 = 445.458$$

2. 計算判定係數 R^2

$$R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} \approx 0.8429$$

(b) 計算誤差變異數的估計值

1. 計算殘差平方和 (SSE) $R^2 = 1 - \frac{SSE}{SST}$

$$SSE = SST \times (1 - R^2)$$

$$= 725.94 \times (1 - 0.7911)$$

$$\approx 151.6489$$

2. 計算 $\hat{\sigma}^2$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{151.6489}{20-2}$$

$$\approx 8.4249$$

(c) 計算判定係數 R^2

• 總平方和 $SST = \sum (y_i - \bar{y})^2 = 631.63$

• 殘差平方和 $SSE = \sum \hat{e}_i^2 = 182.85$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} \approx 0.7105$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- a. the predicted value of y for $x_0 = 4$. $\bar{x} = 1$
- b. the $se(f)$ corresponding to part (a).
- c. a 95% prediction interval for y given $x_0 = 4$.
- d. a 99% prediction interval for y given $x_0 = 4$.
- e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

(a) $x_0 = 4$, 計算 $\hat{y}_0$ $\hat{y}_0 = 1.2 + 0.8 \times (4) = 4.4$

(b) 計算預測標準誤 $se(f)$

1. 計算誤差變異數估計值 $\hat{\sigma}^2$

$$\hat{\sigma}^2 = \frac{\sum \hat{e}^2}{n-2} = \frac{3.6}{5-2} = 1.2$$

2. 計算 $se(f)$

$$se(f) = \sqrt{\hat{\sigma}^2 \left[1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right]}$$

$$se(f) = \sqrt{1.2 \left[1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right]}$$

$$se(f) = \sqrt{1.2 \times (1 + 0.2 + 0.9)} \approx 1.5875$$

(c) 95% 預測區間 $\hat{y}_0 \pm t \times se(f)$ (d) 99% 預測區間

$4.4 \pm 3.1825 \times 1.5875$	$df = n-2 = 3$	$4.4 \pm 5.841 \times 1.5875$
$= 4.4 \pm 5.051$	$t_{0.025, 3} \approx 3.182$	$= 4.4 \pm 9.273$
$[-0.651, 9.451]$	$t_{0.005, 3} \approx 5.841$	$[-4.873, 13.673]$

(e) 當 $x = \bar{x}$ 時, 95% 預測區間比較 當 $x = \bar{x} = 1$ $(x - \bar{x})^2 = 0$

1. $se(f)_{\bar{x}} = \sqrt{1.2 \times (1 + \frac{1}{5} + 0)} = 1.2$

(c) 小題寬度的為 10.1 ($x=4$), 而 (e) 小題的寬度的為 6 ($x=1$), 說明 x 越靠近樣本平均 $\bar{x}$ 時, 預測的精確度越高 (區間越窄)。

2. $\hat{y} = 1.2 + 0.8 \times 1 = 2$

3. 95% 區間 $2 \pm 3.182 \times 1.2 = 2 \pm 3.8184 = [-1.818, 5.818]$